

STREET WASHING EQUIPMENT

KPL 270-200-1250x8



OPERATION INSTRUCTIONS

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SAFETY PRECATIONS



Operating with DYNASET high pressure equipment user has to conform to all safety regulations concerning pressurized equipment. Hydraulic circuit has a high pressure 210 bar.

EQUIPMENT TECHNICAL CONDITION HAS TO PAY SPECIAL ATTENTION.

In order to avoid accidents all operators and maintenance personnel must act in compliance with the laws, regulations and recommendations the high pressure hydraulics, electricity and work safety issued by local authorities.

EXTERME CLEANLINESS MUST BE MAINTAINED WHEN CARRYING OUT ANY SERVICE DISSASSEMBLING OR REPAIR OF HPW-UNIT AND HYDRAULIC SYSTEM. THIS IS CRUCIAL TO ENSURE SAFE, RELIABLE AND LONG-LIFE OPERATION OF YOUR EQUIPMENT.

Hydraulic system has to maintain regularly. Especially couplings, valves and hoses have to be tight. Check and fix possible leakages of pressure and hot blowouts immediately avoiding user and bystander injuries.

The pressure in hydraulic fluid circuit is considerably high what for the technical condition of your equipment should be under constant scrutiny. Especially couplings, valves and hoses should be maintained tight and clean as well as kept under constant observation.

In order to exclude possible accidents, it is not allowed to clean or inspect HPW unit when hydraulic fluid circuit is pressurized. Prior to any cleaning, inspection and service hydraulic system of your carrier machine must be stopped and all hydraulic fluid circuits discharge.

When maintaining and driving the HPW pump in a pump cabinet, wear appropriate clothing and safety equipment such as goggles, ear and eye protection, safety shoes. Beware of machinery parts are warm by hot hydraulic oil.

Operators and maintenance personnel must always comply with local safety regulations and precautions in order to close out the possibility of damages and accidents.

All installation and service of both hydraulic and electric equipment must be performing experienced personnel only.

Preventing freezing nozzles and water circuit, pipeline draining and air flushing have to perform before -0°C temperature.

WARNING STICKER

Dynaset dispatch department includes one (1) warning sticker bag per one (1) main product. Product recipient is obligated to fix determinate warning sticker to Dynaset product. Fix sticker to visible and appropriate place or close to Dynaset product where it's easily seen. Before fixing sticker clean surface with solvent detergent.



READ OPERATOS INSTRUCTIONS.



BEWARE OF HIGH PRESSURE
SPLATTERS.



KEEP FROM FREEZING.



USE EAR PROTECTOR AND
GLOGGLES.

PRODUCT DESCRIPTION

DYNASET KPL 270 - 200 - 1250x8 STREET WASHING EQUIPMENT is especially designed for attached to working machines like mini tractors. KPL street washing unit is ultimately light-weight. KPL high pressure street washer cleans effectively dust, sand and other impurities from street or other fairways. KPL washing efficiency is absolutely superior in comparison to obsolete low pressure street washing equipment. Water consumption is surprisingly low, 1 liter / m².

INSTALLATION

KPL is quick and easy attach to the working machine.

Attaching procedure:

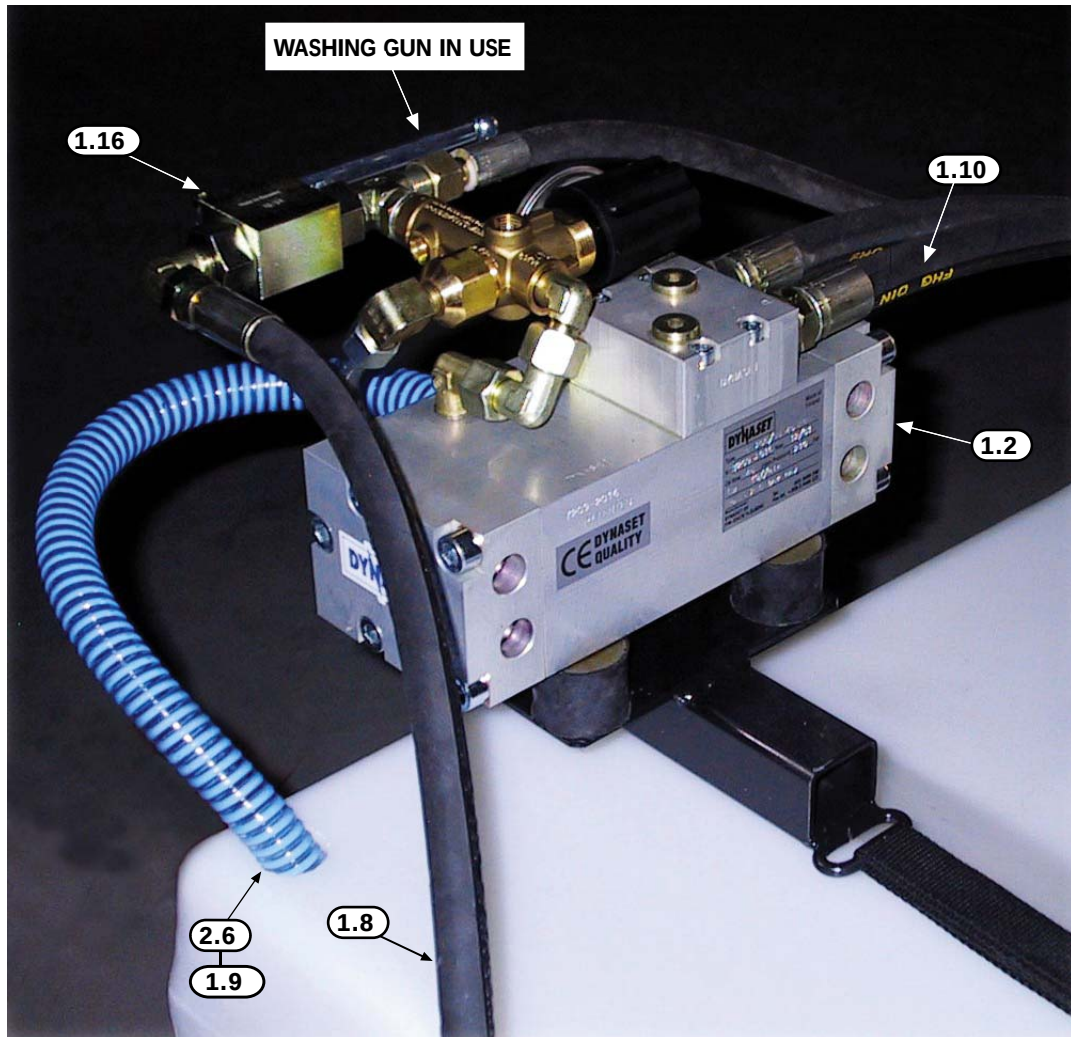
1. Push working machine's mounting frame under the upper hooks (2pc.) in KPL framework.
2. Lift boom up to attach KPL.
3. Fill up the tank with water via cap.
4. Lock the KPL framework to the boom with the tap (lower left framework).
5. Attach the hydraulic hoses power coupling (P, female), tank coupling (T, male) to the working machine.
6. Turn ball valve handle behind the high pressure pump (HPW) to desired position.

View ball valve behind the tank:

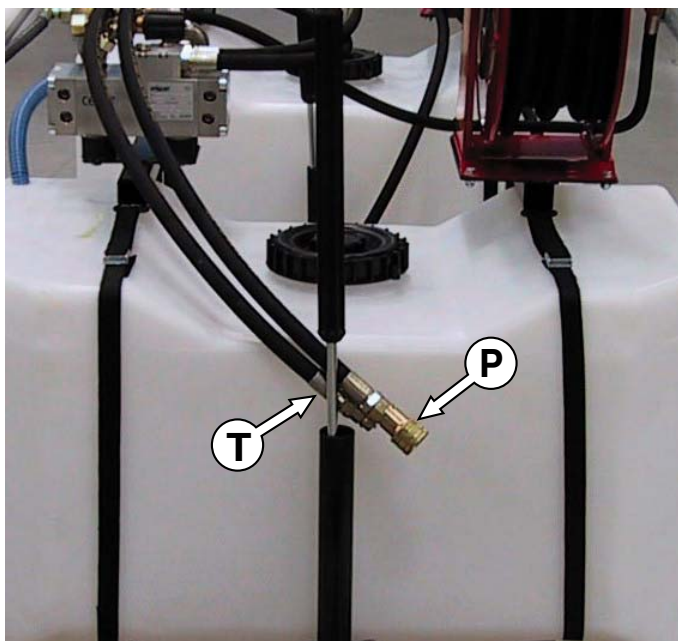
- Handle to the right = Water out of the street nozzle pipe.
 - Handle to the left = Water out of the washing gun.
7. Bend rotate arm (street nozzle pipes) down so that the limiter plate is lowest position.
 8. Switch the supplement hydraulic on and if necessary increase working machine's RPM.



INSTALLATION



HYDRAULIC CONNECTIONS



P = PRESSURE LINE 1/2"

T = TANK LINE 1/2"

KPL PART LIST

Pos	STOCK	DESIGNATION	DETAILS	QUANTITY
1.1	0527116900	KPL270 ASSEMBLY		1
1.2	D200101260	HPW-PUMP, ASSEMBLY	HPW200/30-45	1
1.3	500021653	NOZZLE PIPE	D27 - 1250/8 A2	1
1.4	6306487000	NOZZLE 6501	(3,3 lpm 200 bar 65°)	8
1.5	6306595000	FILTER	R1/8-100 MESH	8
1.6	D210104150	WASHING GUN	MAX.250bar, ST15-SW-800-x	1
1.7	11803493100	REEL HOSE	R3/8-1200/1-R1/2	1
1.8	1803495500	NOZZLE PIPE HOSE	R3/8-2000/1-R3/8 90°	1
1.9	1501064000	SUCTION STRAINER	R3/4 - 30 lpm STR 070/3-M90	1
1.10	1803570000	HYDRAULIC HOSE	R1/2 MALE-1600/2-R1/2 MALE	2
1.11	1702185000	SEAL USIT,	R3/8	5
1.12	1702225000	NIPPLE ADAPTER	R3/8	1
1.13	1702445000	NIPPLE ADAPTER	R3/8u-R3/8s	1
1.14	1701555000	QUICK ADAPTER FEMALE	R1/2	1
1.15	1701560000	QUICK ADAPTER MALE	R1/2	1
1.16	1300876100	BALL VALVE, 3-WAY	R 3/8	1
2.1	0500071167	KPL270 FRAME WORK		1
2.2	0500071168	KPL ROTATE ARM	190, 270	1
2.3	0509010890	FASTENING PLATE	HPW-KPL 200	1
2.4	0500071166	KPL NOZZLE PIPE BRACKET	190, 270	1
2.5	6306736000	HOSE REEL 25m	REWINDING +3/8 R1 HOSE	1
2.6	1803935000	HOSE PVC	3/4 5bar WATER	2
2.7	6306790000	WATER TANK 270L	980x520(350)xh620	1
2.8	1703275000	HOSE COUPLING	25-R3/4 u Ms LVI	1
2.9	8307660000	PIPE BRACKET,	20 STANDARD SERIE 3	2
2.10	8307660200	PIPE BRACKET,	25 STANDARD SERIE 3	2
2.11	1703235000	COUPLER BODY	R3/4, 19, Ms	1

HIGH PRESSURE PUMP (HPW) GENERAL

DYNASET HPW-pump is a converter, which transforms actuating fluid power, usually hydraulic oil, into pumping fluid's power, which is defined by flow rate and pressure. Actuating power is supplied from hydraulic system of carrier machine or other installation. Water or any other pumping fluid can be taken from natural source, reservoir or pressurized supply network.

The patented HPW-pump utilizes the reciprocal motion of hydraulic piston, when two water plungers, flanked to it, develop pressure in delivery (pressure) line. Hydraulic flow moves the piston assembly until the other water plunger reaches its extreme position, when changes the incorporated reversal valve direction of hydraulic flow and, by that, the piston assembly is being set to counter direction. Vacuum is being developed into pumping fluid's intake line and positive pressure in delivery line accordingly. Within pumping cycle water (or other pumping fluid) is being taken by water plungers through intake valves and pumped through pressure valves into delivery line.

Relation of pumping fluid's flow rate and pressure to corresponding parameters of hydraulic fluid is linear, as shown on diagrams below.

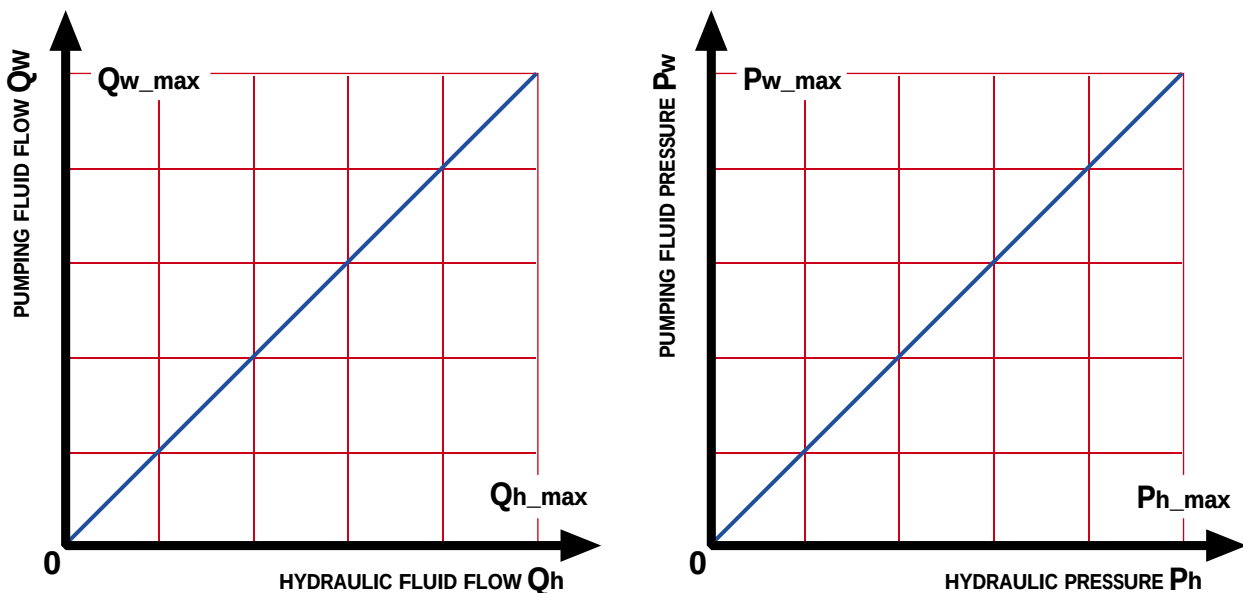
When hydraulic pressure and/or flow is under the value needed to achieve the maximum output power of HPW-pump, the water pressure and flow are decreased respectively without any harm to pump itself.

ATTENTION !

Most of HPW-pumps are self-priming. Specific characteristics of any pump can be verified according to the technical specification (see last page).

THE DESIGN OF HPW-PUMP IS PROTECTED WITH THE INTERNATIONAL PATENT.

PUMPING FLUID TO HYDRAULIC FLUID RELATION IN REGARD TO THEIR FLOW RATES AND PRESSURE.



HPW INSTALLATION

DYNASET HPW-pumps are compact and integrated all-in-one units, designed for both mobile and workshop installation. HPW-pumps perfectly work installed on excavators, mining machines, tractors, trucks and vans. In workshop use HPW-pump is accompanied with hydraulic aggregate.

DYNASET HPW-pump can be positioned deliberately, keeping in view following preconditions:

1.

Easy service access to the unit should be ensured.

2.

Pump should be mounted on sufficiently horizontal surface to enable proper working of leakage detecting bores, which are found either on bottom surface or side faces of pump housing (depends on model). Leakage detecting bores are made to the zones, which separate oil and water chambers from each other.

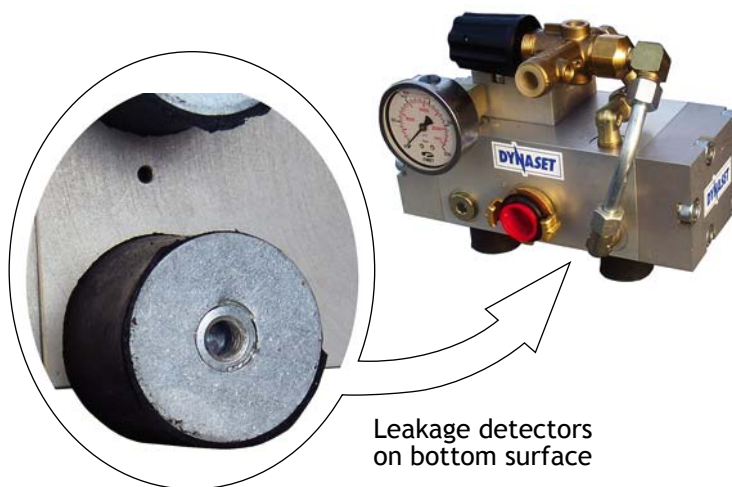
If the draining from leakage detectors exceeds rate of 10 drops per minute, pumps sealing should be replaced. The drain fluid can be water, hydraulic oil or their solution.

You can choose deliberately how to make pump's installation to your carrier: either fixed or as a quick release unit. In latter case HPW-pump is easily detached and re-attached to carrier's hydraulic system with quick couplings, when pump can be moved from one carrier to another when necessary.

... HPW INSTALLATION

LEAKAGE DETECTORS IN DIFFERENT HPW-MODELS

HPW 200/30-45

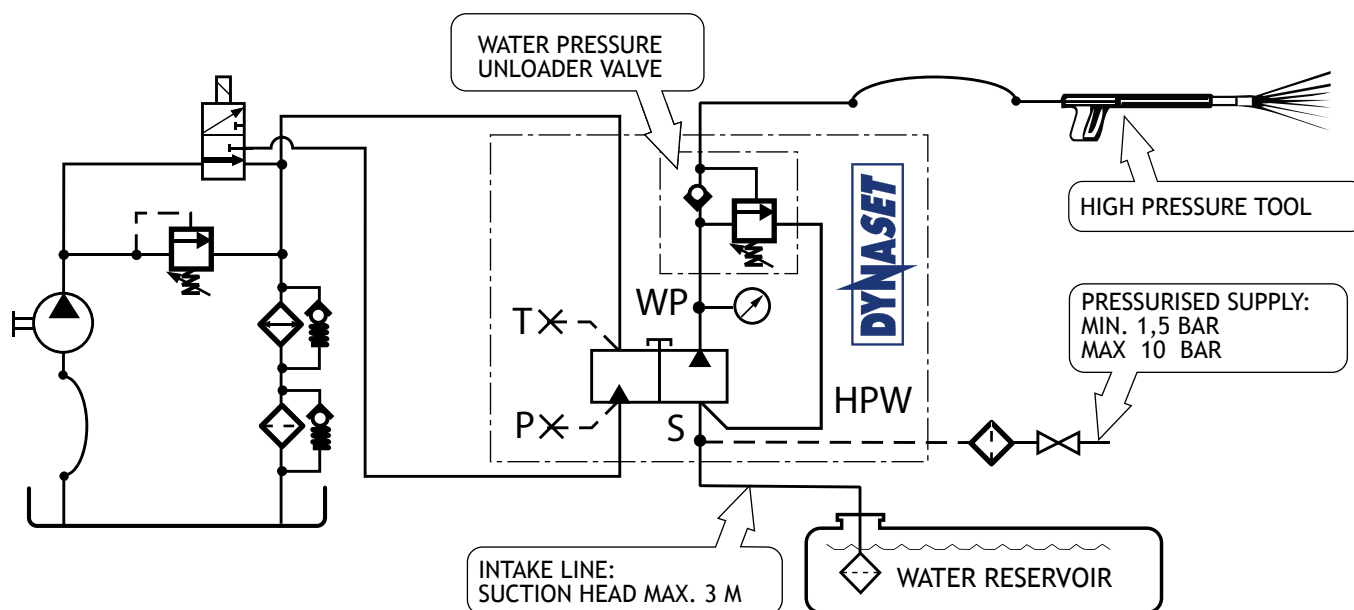


HPW INTEGRATION INTO A HYDRAULIC CIRCUIT

HPW-pump is to be connected to hydraulic system of carrier machine, as well as to pumping fluid circuit. HPW can be installed with no particular efforts to system based on either constant or variable displacement hydraulic pump, as well as to closed hydraulic systems.

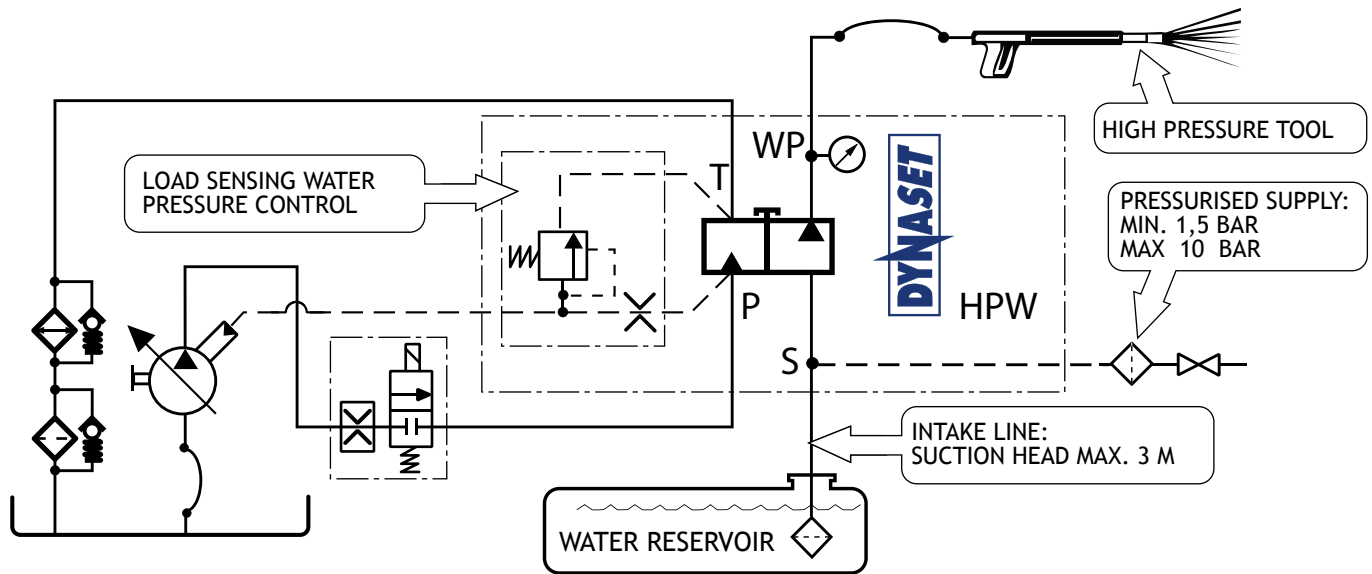
1. **HYDRAULIC SYSTEM WITH CONSTANT DISPLACEMENT PUMP:**
The required oil flow is set by correct dimensioning of the hydraulic pump and keeping the demanded rotation speed or, alternatively, with pressure compensated 3-way valve.
2. **HYDRAULIC SYSTEM WITH VARIABLE DISPLACEMENT PUMP:**
 - .1 In system with OPEN CENTRE the required oil flow is set by **DYNASET PRESSURE COMPENSATED PRIORITY VALVE**
 - .2 In system with CLOSED CENTRE (load sensing system) the required oil flow is set by **DYNASET LS-VALVE** kit, which comprises solenoid valve, pressure compensated oil flow limiter and pressure relief valve.

HPW IN SYSTEM WITH CONSTANT DISPLACEMENT HYDRAULIC PUMP



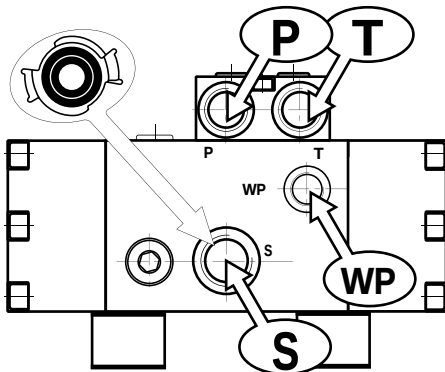
... INSTALLATION

HPW IN SYSTEM WITH VARIABLE DISPLACEMENT HYDRAULIC PUMP



HPW HYDRAULIC PORTS:

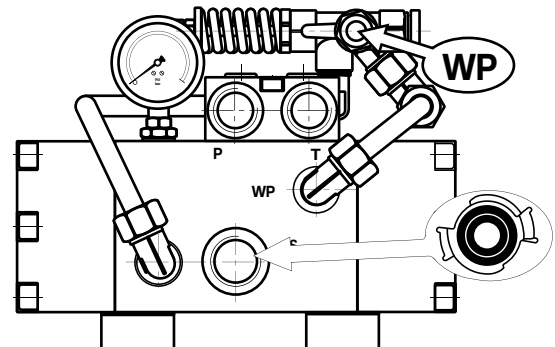
Pressure P, the pressure line is to be attached to;
Return T, the runback line is to be attached to.



HPW-PUMP WITHOUT WATER
PRESSURE UNLOADER VALVE

HPW' WATER PORTS:

INTAKE S, the water supply line is to be attached to;
WATER PRESSURE WP, the delivery (pressure) line
 is to be attached to.



HPW-PUMP EQUIPPED WITH WATER
PRESSURE UNLOADER VALVE

ATTENTION !

Check the ports of your HPW-pump and connect it to hydraulic and pumping fluid circuits according to their markings!

1.

If the hydraulic pressure line will be connected by accident to the return port and return line correspondingly to the pressure port, it will not cause any harm to HPW-pump, however pump will not operate.

2.

Output power of HPW-pump depends on the pressure difference between pressures and return lines. The more is the pressure in return line, the less is concerned pressure difference and output power of the pump. The fact should be taken into account when certain numbers of machines are intended to be joined-up in series.

3.

Since the piston assembly reverses extremely rapidly (refer to the TECHNICAL SPECIFICATIONS), certain oscillation can be imposed to the return line because of pressure pulsation. In order to reduce both pressure pulsation and return line oscillation, the flexible hose with textile fabric cord is recommended to be used in return line to protect from damage low pressure units, such as oil cooler.

4.

Prior to connecting HPW-pump to hydraulic circuit, compatibility of used equipment should be ensured. Hydraulic pressure and flow rate produced by hydraulic circuit should meet values demanded by your HPW-pump. Refer to the technical specifications.

OPERATION

Prior to starting-up the HPW-pump it is checked whether all couplings in hydraulic and pumping fluid circuits are properly connected and there are no leakages over the equipment.
Ensure that the water tank is refilled and intake line's strainer/filter is clean. If water (or any other pumping fluid) is taken from pressurized supply, ensure that the shut-off valve is open.

PUMP' START-UP

HPW-pump starts immediately after the hydraulic flow has been put on either with control valve (manual or solenoid) and/or by pulling the trigger of spray gun.
Immediately after opening the water pressure line, air is being vented out from the pumping fluid circuit and high pressure washing is about to begin!
DO NOT TRY WITH HAND THE WATER OR AIR COMING OUT FROM HIGH PRESSURE NOZZLE!

The nozzle, attached to a pressure tool, determines the flow rate and pressure of water when operating parameters of hydraulic circuit meet HPW-pump's requirements.
In other words, even your HPW-pump fits perfectly to hydraulic system and does well with the job, dimensioning of the pressure tool should never be underestimated as a very important step for ensuring the optimal water jet or hydro-demolition power in each application.

ADJUSTING FLOW RATE AND PRESSURE OF PUMPING FLUID

When the HPW-pump is equipped WITH WATER PRESSURE UNLOADER VALVE:

1.
The valve afore is for adjusting the pressure and flow rate of water or any other pumping fluid. Boosting the pressure increases flow rate and vice versa.
2.
After closing water pressure line, whereupon the adjusted pressure limit is being reached, water pressure unloader valve puts the water to free circulation mode.

When the HPW-pump is installed WITHOUT WATER PRESSURE UNLOADER VALVE:

1.
Pressure and flow in pumping fluid circuit are adjusted by adjusting corresponding parameters in hydraulic circuit.
2.
In circuits based on constant displacement pump closing the water pressure line opens the pressure relief valve in hydraulic system. **IS TO BE KEPT IN MIND THAT HYDRAULIC FLUID CIRCULATION THROUGH PRESSURE RELIEF VALVE CAUSES HEATING OF HYDRAULIC SYSTEM!**
3.
In circuits based on variable displacement hydraulic pump closing the water pressure line adjusts pump's swash plate at 0° angle when hydraulic pump produces flow only for self-lubrication and self-flushing. Pressure control must be fast enough to protect HPW-pump from pressure peaks or the hydraulic system must be provided with pressure peak limiter.

STOPPING HPW-PUMP

HPW-pump is halted by shutting off the hydraulic flow. High pressure remains in closed water delivery line, which should be depressurized to avoid unexpected water discharges. If spray gun or similar tool has been used, pull the trigger for some time after the pump has halted.

In cold season HPW-pump and pumping fluid pressure line should be dewatered after a working shift. Detach the intake hose from HPW-pump and run it for a while until all water is moved from water circuit.

ATTN !

In cold season HPW-pump can be used for circulating the water through water circuit to keep it unfrozen in transit.

Different water based solutions and special liquids can be used as pumping fluid with HPW-pumps. The only requirement is that the pump **MUST** be thoroughly cleaned and flushed with sweet water after the job to prevent pump and its sealing from oxidation and water valves from seizing. The procedure should be made even if such a harmless liquid as sea water has been used.

ATTENTION !

HPW-pump can be run dry unlimited time without any harm, because self-lubricating sealing and circulating hydraulic oil prevent the pump from overheating.

BASIC APPLICATIONS OF HPW-PUMP

Various high pressure washing and cleaning jobs, such as spray gun washing, pipe and sewer cleaning and street washing, are typical applications of DYNASET HPW-pumps. As special applications, high pressure dust suppression, hydro-demolition and high pressure fire-fighting systems are worth to mention. Compact size and reasonable power demand enable to fit HPW-pump practically to any project, where high pressure water or other fluid is being used.

Washing power depends on both water pressure and volumetric flow. High pressure method ensures sufficient water flow at high velocity, offering in that way excellent washing power with exceptionally small water rate in contrast with traditional low pressure method. HIGH PRESSURE SAVES WATER!

HIGH PRESSURE WASHING WITH A SPRAY GUN

Spray gun is a multipurpose high pressure tool, which is delivered either with its own HPW-based power kit for running as stand-alone package, or as optional equipment to another high pressure system.

Spray gun can be fitted with various nozzles and detergent injectors depending on your task.

Three basic nozzle types are shown on the picture alongside:

1

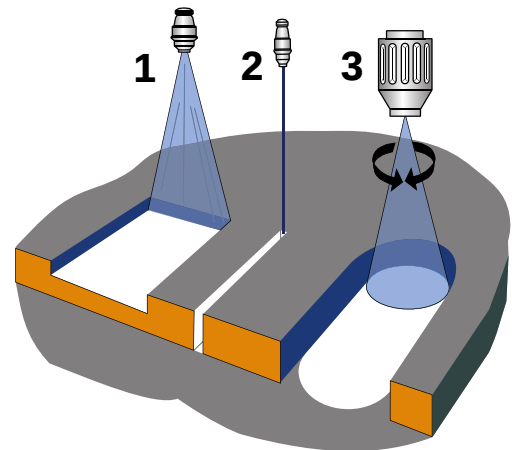
Fan nozzle 15 - 65°;

2

Point nozzle;

3

Rotating point nozzle (turbo-nozzle).



High pressure nozzle is to be chosen according to your washing task.

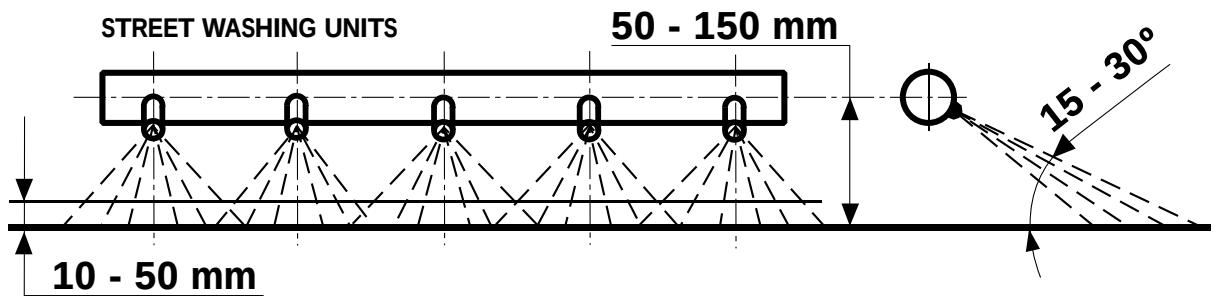
Fan nozzle is to the best advantage when quite big area should be handled in fast and efficient way.

Point nozzle focuses the maximum impact to a small area, fitting perfectly in various hydro-demolitions jobs.

Turbo-nozzle combines advantages of both above nozzle types. It has built-in point nozzle, which is rotated by the recoil force of the side water jet. An impact efficiency of turbo-nozzle is nearly as high as point nozzles and area coverage as big as fan nozzle's.



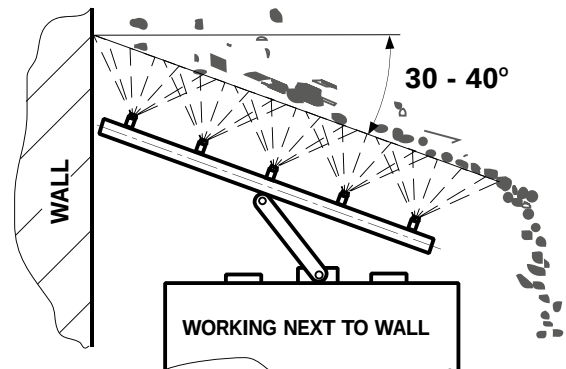
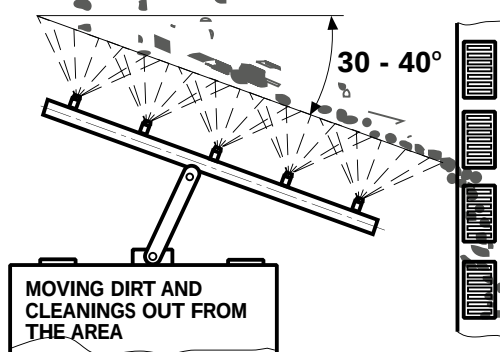
Spray guns for different water outputs, nozzles, high pressure hoses and wide range of accessories are available from DYNASET OY and dealers world-wide.

STREET WASHING UNIT

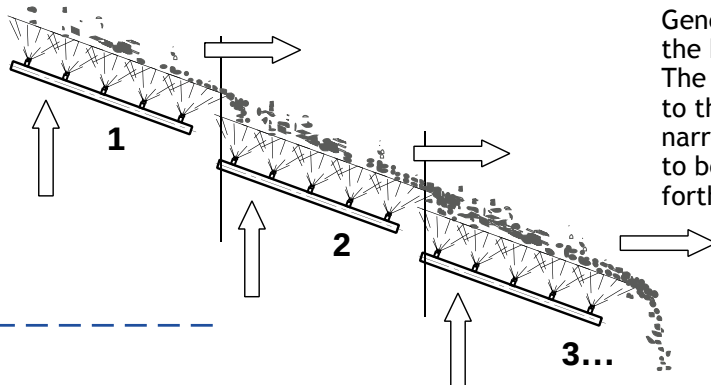
When adjusting on-position, the street washing pipe should be brought down horizontally as close as possible to the roadway. An operator should be careful, though most KPL-types are provided with support wheel in order to prevent street washing pipe from damage. Street washing pipe should be positioned in relation to surface to be washed so, that each water jet would meet its adjoins at a distance of 10–50 mm from it (refer to the drawing above).

The kind of paving and surface unevenness should be taken into account as well.

It is kept in mind, that positioning the street washing pipe to high would cause washing power loss up to 50 %.



The steering angle of street washing pipe is determined by direction, where the dirt is being moved to. The angle of 30–40° is proved to be most effective - refer to the above sketches. High pressure water jets work as dozer blade, moving dirt and cleanings along the street washing pipe to its other end and further aside.



Generally, it is recommended to start job from the highest point of area to be washed. The dirt and cleanings are to be run directly in to the waste water channels when cleaning narrow lanes, or ridged at the driving lane side to be moved further at next driving and so forth, until the whole area will be cleaned.

MAINTENANCE

KPL:

Clean suction strainer from impurities at least once a year or if needed. Access to suction strainer happens via tank cap.

Remove nozzle if clogged, clean the strainer and nozzle from dirty with clean water. Change new Teflon tape onto the threads and fasten nozzle.

Put grease into grease nipples at rotate arm joint once a year.

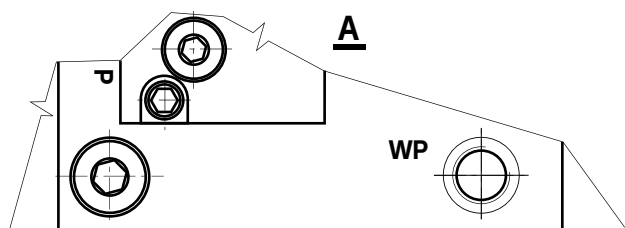
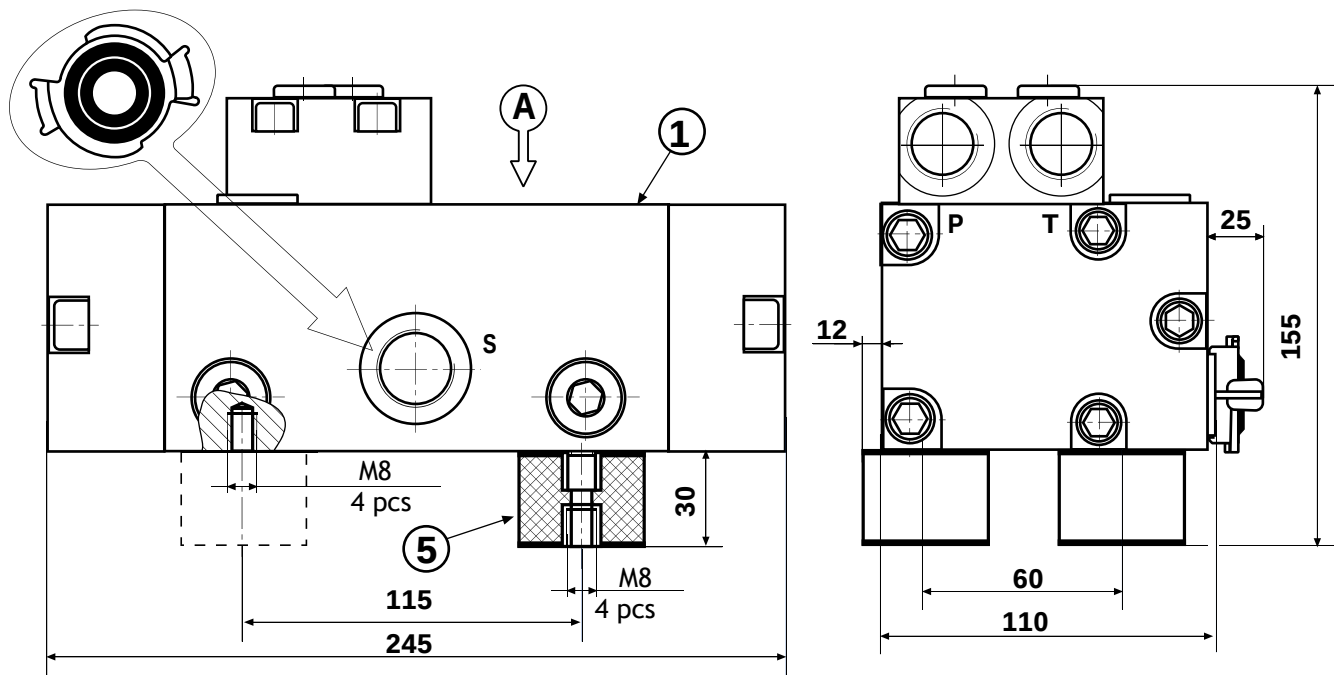
HPW:

Due to small number of moving parts, in normal operating conditions HPW-pump does not require any other service, except replacement of sealing or occasionally of water valves, which depends on content of abrasives in pumping fluid, as well as on cleanliness of hydraulic oil.

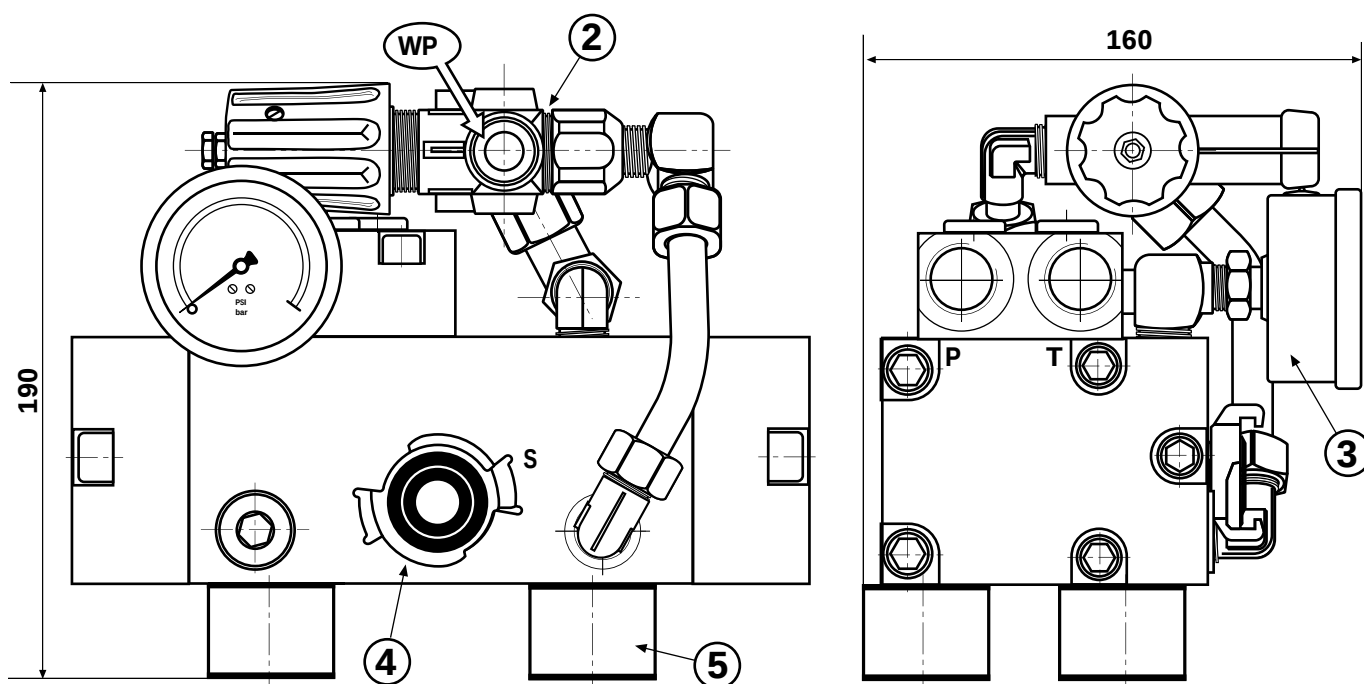
Check constantly whether the fluid dropping from pump's leakage detectors is growing and replace sealing in proper time to exclude intermixing of hydraulic oil and pumping fluid.

To enable the easy surveillance, pump should be kept clean.

WHEN CARRYING OUT ANY SERVICE DISSASSEMBLING OR REPAIR OF HPW-PUMP, ABSOLUTE CLEANLINESS MUST BE MAINTAINED TO ENSURE RELIABLE AND TROUBLE-FREE OPERATION OF YOUR EQUIPMENT.

HPW OUTLINE DIMENSIONS**HIGH PRESSURE PUMP HPW 200/30-45-ST****FLUID CONNECTIONS**

HYDRAULIC PORTS		PUMPING FLUID PORTS	
pressure P	return T	pressure WP	intake S
BSP 1/2"	BSP 1/2"	BSP 3/8"	BSP 3/4"

HIGH PRESSURE PUMP HPW 200/30-45-ST

HPW TECHNICAL CHARACTERISTICS

PARAMETERS		HPW 200/30-45	
WATER POWER			
WATER FLOW	max	l/min	30
WATER PRESSURE	max	bar	200
WATER TEMPERATURE	max	°C	70 °C
PUMPING POWER	max	kW	10
SUCTION HEAD	max	m	3
WATER FEED PRESSURE	min-max	bar	-0,3 ...10
PRESSURE RATIO	water/hydraulic		1,18
SPEED RATIO		1/min	750
WATER PISTONS	outer diameter	mm	D 30
WATER CONNECTIONS			
PRESSURE LINE			BSP 3/8"
SUCTION / FEED LINE	min		BSP 3/4"
PRESSURE HOSE	recommendation		3/8"
SUCTION HOSE	recommendation		3/4"
WATER FILTER	min.	mesh	80
HYDRAULIC CONNECTIONS			
PRESSURE LINE		P	BSP 1/2"
RETURN LINE		T	BSP 1/2"
SERVICE LINE		S	BSP 1/4"
HYDRAULIC FLUID REQUIREMENTS			
VISCOSITY		cSt	10 - 200 / optimal 25 - 35
TEMPERATURE		°C	max. 70
FILTER RATIO	recommendation	um	min. 25
COOLING CAPACITY	recommendation	kW	2
HYDRAULIC POWER REQUIREMENTS			
HYDRAULIC FLUID FLOW		l/min	45
OPERATING PRESSURE	Δp	bar	185
MAXIMUM PRESSURE		bar	210
IDLE RUN PRESSURE		bar	10
RETURN LINE PRESSURE	max	bar	210
WEIGHT		kg	8
PRESSURE GAUGE			OPTIONAL
WATER PRESSURE UNLOADER	ST		OPTIONAL
LS-VALVE	LS		OPTIONAL

HPW PART / COMPONENT LIST

Pos	STOCK	DESIGNATION	DETAILS	QUANTITY
	NUMBER			HPW200/30-45
1	D200101260	HPW-PUMP, ASSEMBLY	HPW200/30-45	
1.1	0400001737	CENTRE BODY, ASSEMBLY		1
1.1.1	0400000221	BLOCK 63	HPW 200	1
1.1.2	1702705000	HF-PLUG	R 3/8"	3 (2, 1)
1.3	0400001122	P/T BLOCK, ASSEMBLY		1
1.3.1	0400000222	BODY	R 1/2"	1
1.3.2	1702695000	HF-PLUG	R 1/4"	2
1.3.3	2204084000	SEAL	Included into a seal kit	2
1.3.4	8500409800	CAPSCREW, SOCKET HEAD	M6x40 8.8	4
1.4	0400001741	WATER HEAD, LEFT, ASSEMBLY		1
1.4.1	0400030820	HEAD BODY, LEFT	L 30	1
1.4.2	2104524000	WATER PISTON SEAL, 2T	Included into a seal kit	1
1.4.3	2204094000	SEAL	Included into a seal kit	2
1.4.4	1702710000	PLUG, SOCKET HEAD	RK3/8"	2
1.5	0400001028	WATER HEAD, RIGHT, ASSEMBLY		1
1.5.1	0400000219	HEAD BODY, RIGHT	R 30	1
1.4.2	2104524000	WATER PISTON SEAL, 2T	Included into a seal kit	1
1.4.3	2204094000	SEAL	Included into a seal kit	2
1.4.4	1702710000	PLUG, SOCKET HEAD	RK3/8"	2
1.6	0400001027	SEALING FLANGE	HPW 200/30-45	2
1.6.1	0400030218	FLANGE BODY		2
1.6.2	2104514000	WATER PISTON SEAL, 1T	Included into a seal kit	2
1.4.2	2104524000	WATER PISTON SEAL, 2T	Included into a seal kit	2
1.6.3	2204304000	FLANGE SEAL	Included into a seal kit	2
1.7	0400009491	INTAKE VALVE	D25-22	2
1.7.1	6106370000	VALVE SEAT		2
1.7.2		VALVE DISK		2
1.7.3		VALVE CAP		2
1.7.4	0409013076	SPRING, SUCTION		2
1.7.5	2204094000	WATER VALVE SEAL	Included into a seal kit	2
1.8	0400009492	DISCHARGE VALVE	D25-22	2
1.7.1	6106370000	VALVE SEAT		2
1.7.2		VALVE DISK		2
1.7.3		VALVE CAP		2
1.8.4	0400116851	SPRING, PRESSURE		2
1.7.5	2204149000	WATER VALVE SEAL	Included into a seal kit	2
1.9	0400001026	PISTON ASSEMBLY	HPW200	1
1.9.1	0400000886	HYDRAULIC PISTON	55/44,3	1
1.9.1.2	8500679100	SET SCREW	SOCKET HEAD M5x10	2
1.9.1.3	2104639500	SEAL, 2T	Included into a seal kit	1
1.9.2		REVERSAL VALVE		1
1.9.2.1	0400021679	VALVE SPOOL		1
1.9.2.2	0400021680	VALVE SEAT		1
1.9.3	0400030887	WATER PISTON		2
1.9.4	0400001062	REVERSAL VALVE GUIDE		1
1.9.4.1	2104594000	SEAL, 2T	EXCLUDED FROM A SEAL KIT	1
1.9.5	0400000217	CYLINDER LINER	44,3	1
1.9.5.1	2204304000	SEAL 57,00x3,00	Included into a seal kit	1
1.10	8500415600	CAPSCREW, SOCKET HEAD	M10x45 8.8	10
2	V100102800	UNLOADER VALVE, COMPLETE	DST-261 /OPTIONAL (-ST)	1
3	1501302000	PRESSURE GAUGE, COMPLETE	D63, 250 bar, 1/4	1
4	1703275000	HOSE COUPLING	25- R 3/4"	1
5	8307566000	RUBBER CUSHION	40/30-M8	4

HPW PART / COMPONENT LIST

2. UNLOADER VALVE ASSEMBLY Code V100102800				
2.1	6306376000	UNLOADER VALVE	DST-261	1
	1702745000	PLUG, SOCKET HEAD	RK1/4"	1
2.2	1701980000	ELBOW FITTING	12L RK 3/8"	1
2.3	1702105000	SWIVEL ELBOW FITTING	12L	1
2.4	0400001386	ELBOW FITTING	RK 3/8 P12	1
2.5	0400001282	BY-PASS PIPE, COMPLETE	D10 108	1
2.6	1701960000	ELBOW FITTING	10L RK 3/8"	1
2.7	1701945000	ELBOW FITTING	10L RK 1/4"	1
3	1501302000	PRESSURE GAUGE , COMPLETE		1
3.1	170202xxxx	ELBOW FITTING	NPT1/4"-RK3/8"	1
SEAL KIT Code V100101155				
1.3.3	2204084000	SEAL		2
1.4.2	2104524000	WATER PISTON SEAL, 2T		4
1.4.3	2204094000	SEAL		4
1.6.2	2104514000	WATER PISTON SEAL, 1T		2
1.6.3	2204304000	SEALING FLANGE SEAL		2
1.7.5	2204149000	WATER VALVE SEAL		4
1.9.1.3	2104639500	SEAL, 2T		1
1.9.5.1	2204304000	SEAL		1
WATER VALVE KIT Code V100101235				
1.7	0400009491	INTAKE WATER VALVE		2
1.8	0400009492	DISCHARGE WATER VALVE		2
1.7.5	2204149000	WATER VALVE SEAL	Incl. in valve assemblies (2x2)	4
1.4.3	2204094000	SEAL		4
SCREW KIT Code V100101315				
1.3.4	8500409800	SCREW	M6x40 8.8	4
1.10	8500415600	SCREW	M10x45 8.8	10

HYDRAULIC FLUIDS

Wide range of standard hydraulic fluids can be used with the DYNASET hydraulic equipment.

Subject to the operating temperature, following mineral hydraulic oils are recommended:

ISO VG 32	for oil's operation temperature up to 70 °C;
ISO VG 46	for oil's operation temperature up to 80 °C;
ISO VG 68	for oil's operation temperature up to 90 °C.

Synthetic and bio-oils can be used as well if their viscosity characteristics and lubricating efficiency are corresponding to above mineral oils. Automatic transmission fluids and even engine oils can be used, provided that they are allowed to be used in hydraulic system of your carrier machine.

Prior to use special hydraulic fluids a with DYNASET equipment, please be kindly requested to contact nearest DYNASET representative for an advice.

PUMPING FLUIDS

Most of liquids, which can be transferred by pumping, can be pumped also with HPW-pump.

Water based fluids and sea water can be pumped without any special arrangements, however the pump **MUST BE FLUSHED AND CLEANED THOROUGHLY** after a working shift with clean sweet water. Same rule applies to pumping of gelatin and depositing fluids –after having done your job, flush HPW-pump with clean water or suitable solvent.

Solvents, fuels and oils can be pumped without any special arrangements.

Fluids with abrasives or high content of solids can be pumped as well, but should be kept in mind, that HPW-pump's service life time is considerably shortened. Pls. note that DYNASET HDF-pump series is especially designed for pumping of abrasive fluids.

When HPW-pump is going to be used as a self-priming unit with a special or high viscosity liquid, the flow and self-priming conditions should be evaluated. If the self-priming property can not be reached, pumping liquid is to be fed with certain pressure (refer to the TECHNICAL SPECIFICATIONS), or the pump should be used at lower hydraulic flow.

Furthermore, if pumping solution is aggressive to NBR-rubber, special seals must be used.

HPW-models made of special materials or with coated parts are available by request.

PRODUCT DISPOSAL

Conform to waste legislation, regulations and recommendations in waste disposal and waste recycling issued by local authorities.

Precondition:

- Product is put properly out of order.
- Before transportation get off all used agent (oil, cooling liquid) and dirty filters.

Items requiring specialist treatment:

- Separate the base materials, iron, copper, steel, electronic pcbs, removing paint, polyester resin, and insulation tape and/or plastics residues from all components.

Dispose of this 'waste material' can now be recycled.

Forward the reclaimed materials for recycling.

Waste material:

- Dispose of waste material from both of the above processes via an authorized disposal company.

Customer can send us the old Dynaset equipment for reuse or recycling at the purchaser's cost, adequately packed for shipment, to the DYNASET OY or to other location authored by DYNASET OY.

Shipment documents must contain purchaser's name and contact information.

HPW TROUBLESHOOTING**1. HPW-PUMP DOES NOT PRODUCE WATER FLOW**

1.1 HPW-PUMP DOES NOT WORK	1.1.1 Too small hydraulic flow or no hydraulic flow at all. 1.1.2 Hydraulic pressure too low. 1.1.3 Hydraulic flow reversed. 1.1.4 Hydraulic piston damaged mechanically.	1.1.1 Enable or adjust the hydraulic flow. 1.1.2 Adjust the hydraulic pressure. 1.1.3 Check and reconnect hydraulic hoses. Pressure hose should be connected to P-port and return hose to T-port. 1.1.4 Replace damaged part.
1.2 HPW-PUMP WORKS, BUT DOES NOT DELIVER WATER FLOW. <i>Models with water pressure unloader valve.</i> →	1.2.1 Water intake and pressure valves are open (jammed with debris) or damaged. 1.2.2 Regulator's unloader valve opens from intake to pressure.	1.2.1 Check water valves and clean them thoroughly or replace when damaged. 1.2.2 Check the valve and repair failure.
1.3 HPW-PUMP DOES NOT RECEIVE PUMPING FLUID	1.3.1 Intake hose detached or hose breathes. 1.3.2 Water supply line clogged. 1.3.3 Suction head too high.	1.3.1 Check and fix the hose and connectors. 1.3.2 Check strainer or water filter and clean thoroughly. 1.3.3 Check the performance with pressurized water supply when possible.
1.4 WATER PRESSURE LINE BLOCKED. <i>Models with water pressure unloader valve.</i> →	1.4.1 Nozzle clogged. 1.4.2 Regulator's check valve damaged, pressure line blocked.	1.4.1 Check the nozzle and clean thoroughly. 1.4.2 Check the valve and repair failure.

HPW TROUBLESHOOTING

2 WATER FLOW RATE TOO SMALL	2.1 Insufficient hydraulic flow or pressure. 2.2 Nozzle of your pressure tool undersized. 2.3 Pressure loss in delivery hose.	2.1 Adjust the hydraulic flow to the demanded level at required pressure. 2.2 Verify the nozzle sizing and replace with proper one. 2.3 Verify the hose sizing and replace with proper one.
3. WATER PRESSURE TOO LOW	3.1 Insufficient hydraulic pressure or flow. 3.2 Nozzle of pressure tool oversized. 3.3 Wear-out of nozzle. 3.4 Insufficient water supply. 3.5 Water pressure unloader valve set too low.	3.1 Adjust the hydraulic pressure to the demanded level at required hydraulic oil flow. Pressure loss should be minimized - do not use hoses of too small diameter or/and of an excessive length. 3.2 Verify the nozzle sizing and replace with a proper one. 3.3 Replace the nozzle. 3.4 Check and fix the problem. Use pressurised water supply if available. 3.5 Check and re-adjust to specification.
4. INTENCE PULSATING OF WATER PRESSURE	4.1 Some of water intake and pressure valves are open or damaged. 4.2 Water intake line breathes causing pump cavitation. 4.3 Water intake line's diameter too small, resulting in pump cavitation. 4.4 Excessive hydraulic flow, when HPW-pump runs too fast.	4.1 Check water valves, clean thoroughly or repair. 4.2 Check water intake line and fix the problem. 4.3 Verify the hose sizing and replace with proper one. 4.4 Adjust the hydraulic flow to the demanded level.

HPW TROUBLESHOOTING

5. CLOSING THE WATER PRESSURE LINE DOES NOT DROP PRESSURE TO FREE CIRCULATION MODE. Hydraulic system starts to run through the pressure relief valve, emitting excessive heat. <i>Models with water pressure unloader valve.</i> →	5.1 Insufficient hydraulic pressure in relation to pumping fluid pressure, adjusted with water pressure unloader valve. 5.2 Defective water pressure unloader valve.	5.1 Adjust the hydraulic pressure up as much as necessary to enable proper operation of water pressure unloader valve. Note that the maximum hydraulic pressure should not be overrun ! If boosting the hydraulic pressure is not possible, the water unloader pressure setting should be dropped. 5.2 Repair or replace water pressure unloader valve.
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6. LEAKAGES	6.1 Hydraulic oil leakages. 6.2 Pumping fluid leakages. 6.3 Hydraulic-pumping fluid commixture dropping from leakage detecting bores.	6.1 Check the tightness of component mating, tighten screws. Replace when necessary sealings of pump's mated surfaces. Check and tighten/replace couplings. 6.2 Check the tightness of component mating, tighten screws. Replace when necessary sealings of pump's mated surfaces. Check and tighten/replace couplings. 6.3 If the draining from leakage detectors exceeds rate of 10 drops per minute, pump's sealings should be replaced. The rule applies to dropping of hydraulic fluid, pumping fluid and their commixture.
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MANUFACTURER'S LIMITED WARRANTY

1. Warranty coverage

All hydraulic accessories manufactured by DYNASET OY are subject to the terms and conditions of this limited warranty. Products are warranted to the original purchaser to be free from defects in materials or workmanship. Exclusions from warranty are explained in item 8.

2. Beginning of warranty period

Warranty period begins from the delivery date of the product. Delivery is considered to be done on the date when installation has been accomplished or purchaser has taken the product in use. Product is considered as taken in use at the date when DYNASET OY has delivered the product to purchaser, unless separately agreed otherwise by written agreement.

3. Warranty period

Warranty period is twelve (12) months based on maximum of 2000 hours usage during this time period. In cases where the system is provided complete with certain special components (e.g. drive unit), those components are considered as a subject to their manufacturer's warranty.

4. Warranty procedures

Immediately upon identifying a problem which purchaser believes to be a failure subject to the product's limited warranty, purchaser must contact primary to the seller of the product. Contact must be made as soon as possible, latest thirty (30) days after the problem was identified. Seller and/or manufacturer technical staff determines the nature of the problem primarily by phone or e-mail. Purchaser commits to give necessary information and to perform routine diagnostic procedures in order to determine the nature of the problem and necessary procedures.

5. Warranty repairs

If the product is found to be defective during the warranty period, DYNASET OY will, at its option, either repair the product, author it to be repaired at its authorized workshop or exchange the defective product. If the product must be repaired elsewhere than premises of DYNASET OY or authorized workshop, all costs excluded from this warranty (traveling and waiting hours, daily allowance, traveling expenses and uninstallation/reinstallation costs) will be charged from the purchaser.

If the problem is not covered by this limited warranty, DYNASET OY has the right to charge purchaser of troubleshooting and repairing.

6. Delivery terms of warranty repair

If the product is found possible to be defective under this limited warranty and it needs to be repaired, DYNASET OY gives Warranty Return Number (WRN). Items being returned must be shipped, at the purchaser's cost, adequately packed for shipment, to the DYNASET OY or to other location authored by DYNASET OY.

Shipment documents must contain:

- Purchaser's name and contact information
- Receipt of original purchase
- WRN code
- Problem description

7. Warranty of repaired product

Warranty period of the product repaired under this limited warranty continues to the end of original warranty period.

8. Exclusions from warranty

This warranty shall not apply to:

- a. Failures due to normal wear and tear, improper installation, misuse, abuse, negligence, purchaser selection of improper product to intended use, accident, improper filtration of hydraulic oil or intake water or lack of maintenance
- b. Cost of maintenance, adjustments, installation or startup
- c. Coating, hydraulic oil, quick couplings and interconnection hoses (internal or external to system assemblies)
- d. Products altered or modified in a manner not authorized by DYNASET OY in writing
- e. Products which have been repaired during warranty period by others than DYNASET OY or its authorized workshop
- f. Costs of any other damage or loss, whether direct, indirect, incidental, special or consequential, arising out of the use of, or the inability to use, the product
- g. Telephone or other communications expense
- h. Product that is used in exceptional conditions, considered to cause excessive wear and tear
- i. Faults caused by nature phenomenon's like flood, thunder, etc.

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EC DECLARATION OF CONFORMITY

We hereby declare that the below-reference product design, structure and our manufacture form fulfills relevant, fundamental safety and health requirements with the provisions in the Council Directive on mutual approximation of laws of the Member States on the safety of machines.

- Machine directive 2006/42/EC
- Directive 97/23/EC European standards Pressure equipment.

Apply conformity standards:

- CEN EN ISO 4413: EN ISO 4413:2010
Hydraulic fluid power - General rules and safety requirements for systems and their components.

If device has changed by someone other than at the hands of manufacturer or his permission, this declaration is not valid.

PRODUCT: STREET WASHING EQUIPMENT

TYPE NUMBER: KPL 270 - 200 - 1250x8

YLÖJÄRVI 11.05.2012

DYNASET Oy



Timo Nieminen
R&D Manager

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